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## H2 Biology

9744/03

Paper 3 Long Structured and Free-response Questions

20 September 2021

2 hours

Candidates answer **Section A** on the Question Paper and **Section B** on the separate Answer Booklet.

Additional Materials: 12-page Answer Booklet.

### READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

#### Section A

Answer **all** questions.

#### Section B

Answer **one** question.

Write your answer to each part of the question on a fresh sheet of paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

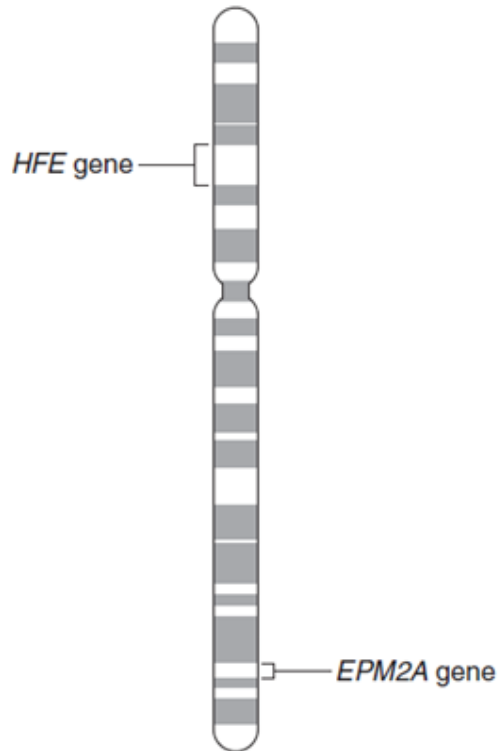
| For Examiner's Use |    |
|--------------------|----|
| Section A          |    |
| 1                  |    |
| 2                  |    |
| 3                  |    |
| Section B          |    |
| 4 OR 5             |    |
| Total              | 75 |

This document consists of **19** printed pages and **1** blank page.

**Section A**

Answer **all** the questions in this section.

- 1** Fig. 1.1 shows the location of two genes, *HFE* and *EPM2A*, on human chromosome 6.

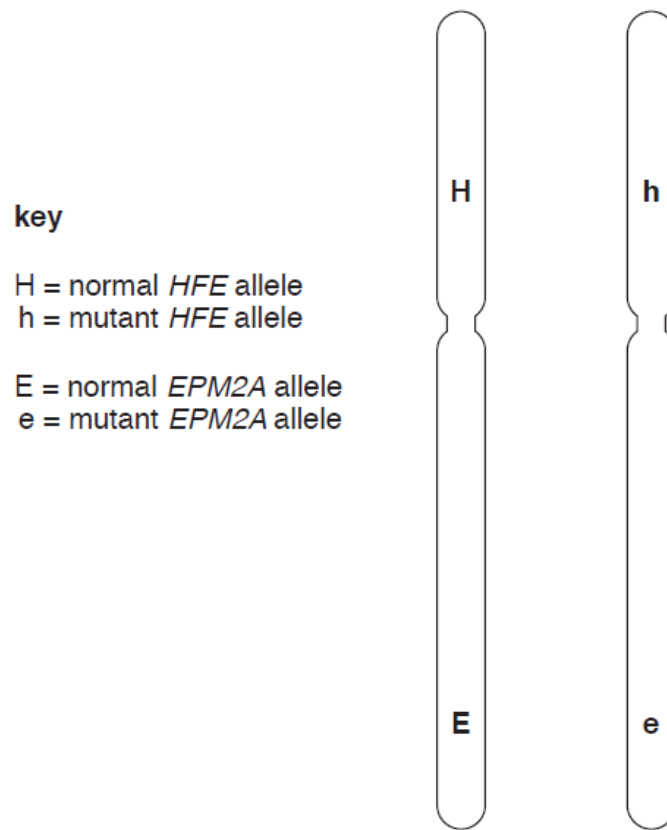


**Fig. 1.1**

The protein product of the *HFE* gene is involved in the absorption of iron. A mutation of the *HFE* gene, where a cysteine is replaced by tyrosine at position 282, can lead to the disorder hereditary haemochromatosis.

Laforin, the protein product of the *EPM2A* gene, has a number of roles in the body. A deletion mutation of the *EPM2A* gene has been linked to Lafora disease.

Fig. 1.2 shows the genotype of a healthy human female, Mary.



**Fig. 1.2**

Mary's husband has both hereditary haemochromatosis and Lafora disease. They have two children:

- a daughter, who is healthy
- a son, who has both hereditary haemochromatosis and Lafora disease.

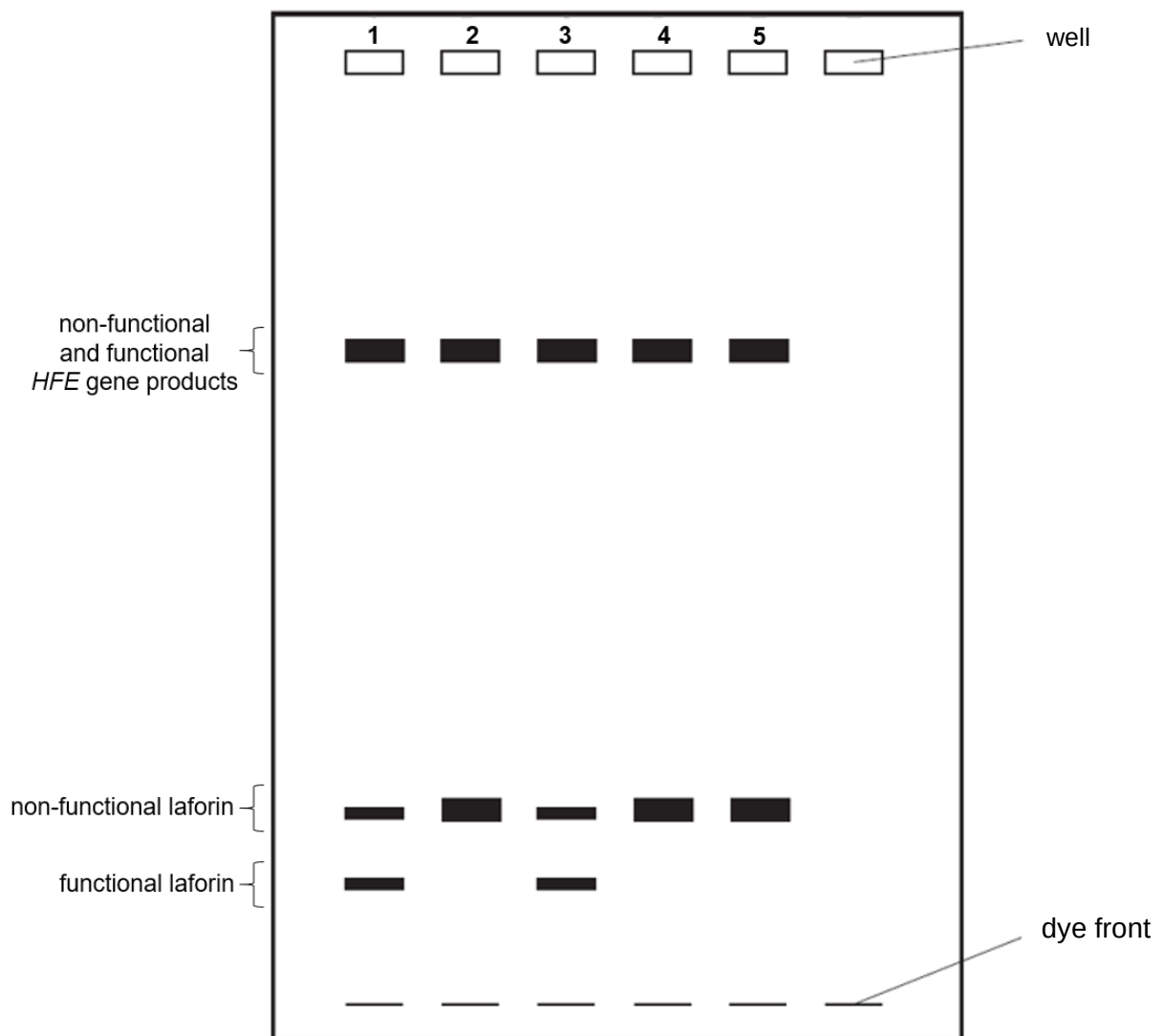
- (a) Draw a genetic diagram to explain the cross between Mary and her husband.

The technique of polyacrylamide gel electrophoresis (PAGE) is used to separate and identify proteins. One method of PAGE involves treating proteins with a detergent, such as sodium dodecyl sulfate (SDS), to dissociate proteins into their constituent polypeptide subunits. Proteins treated with SDS have a uniform net charge on each polypeptide so that during electrophoresis they are separated only on the basis of their relative molecular mass.

After treatment with SDS, proteins are placed in wells cut into a polyacrylamide gel. A dye is added to each sample to show the progress of the samples across the gel. A current is applied to the gel and when the dye reaches a point towards the end of the gel, the current is switched off. The relative mobility of each polypeptide is calculated as follows:

$$\frac{\text{distance travelled by polypeptide band}}{\text{distance travelled by dye front}}$$

Gene products of the *HFE* gene and *EPM2A* gene from Mary and her family were analysed with SDS-PAGE and the results are shown in Fig. 1.3.



**Legend:**

- Lane 1 – Mary
- Lane 2 – Mary's husband
- Lane 3 – Mary's daughter
- Lane 4 – Mary's son
- Lane 5 – Individual X

**Fig. 1.3**

- (b) (i) The distance travelled by the polypeptide band of both the non-functional and functional *HFE* gene products is 38 mm, while that travelled by the dye front is 128 mm. The relative mobility of non-functional and functional *HFE* gene products is 0.297.

Calculate the relative mobility of non-functional and functional laforin, and fill in Table 1.1 with your calculated values.

**Table 1.1**

| protein                                | relative molecular mass | relative mobility |
|----------------------------------------|-------------------------|-------------------|
| non-functional <i>HFE</i> gene product | 68000                   | 0.297             |
| functional <i>HFE</i> gene product     | 68000                   | 0.297             |
| non-functional laforin                 | 34000                   |                   |
| functional laforin                     | 29000                   |                   |

[1]

- (ii) Using the information provided, explain the relative mobility of the non-functional and functional *HFE* gene products.

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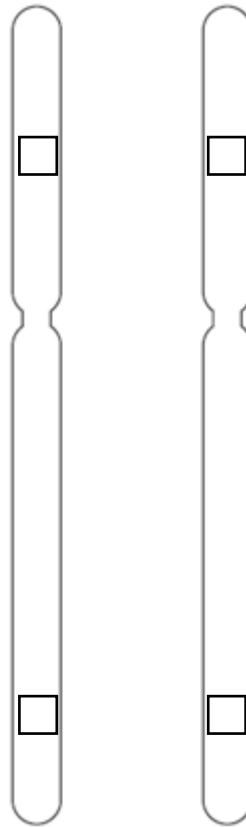
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..... [2]

- (iii) Individual X is the third child of Mary and her husband, and suffers from Lafora disease but not hereditary haemochromatosis. Fig. 1.4 shows a pair of chromosome 6 in individual X.

On Fig. 1.4, indicate the genotype of individual X in the boxes.



**Fig. 1.4**

[1]

- (iv) On Fig. 1.4, circle the chromosome that individual X inherited from Mary. [1]

- (v) Using your knowledge on gamete formation, describe how individual X came to inherit the combination of alleles indicated in Fig. 1.4 from Mary.

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- (c) Suggest if genetic diseases such as Lafora disease is an example of continuous or discontinuous variation.

Provide **two** reasons to support your answer.

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.....[3]

- (d) Mary and her family reside in Australia where the giant moray eel can be found. The sizes of eel populations tend to remain relatively stable despite eels producing large numbers of offspring.

Suggest **two** reasons why the population sizes of eels tend to remain relatively stable.

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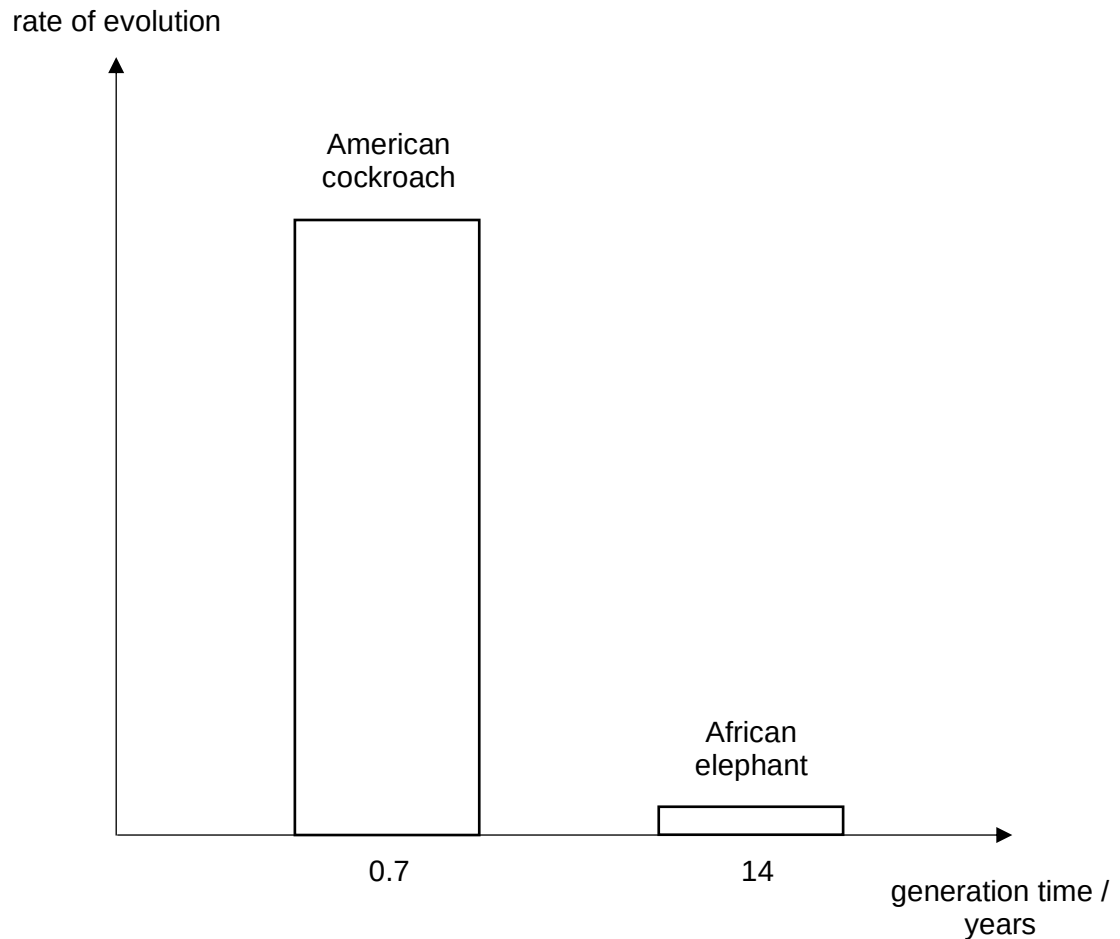
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..... [2]



The generation time of a species is the average number of years between the birth of one generation (parents) and the birth of the next generation (offspring). Fig. 1.5 shows the generation time of the American cockroach and the African elephant.



**Fig. 1.5**

- (e) One of the most important factors in determining the rate of evolution is the introduction of new alleles within a species.

Describe and explain the relationship shown in Fig. 1.5.

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.....[3]

The habitat of the giant moray eel includes coral reefs as shown in Fig. 1.6.



**Fig. 1.6**

Corals grow in shallow sea water. Corals consist of colonies of small animals called polyps. These polyps have photosynthetic algae living within their cells, and this relationship is advantageous to both the coral polyps and to the algae.

The algae that live within the cells of coral polyps can also live independently as free-living algae in the sea water.

- (f) Suggest why the rate of photosynthesis of algae that live within the cells of coral polyps is higher than that of free-living algae.

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.....[3]

The algae that live within the cells of coral polyps have five different chloroplast pigments.

Table 1.2 shows the wavelengths of light with the largest absorption for each algae chloroplast pigment.

**Table 1.2**

| chloroplast pigment        | wavelength / nm |
|----------------------------|-----------------|
| chlorophyll a              | 430             |
| peridinin                  | 456             |
| chlorophyll c <sub>2</sub> | 450             |
| dinoxanthin                | 442             |
| β-carotene                 | 454             |

Corals can be kept in glass tanks that are usually lit by lamps radiating mainly violet and blue light, with wavelengths in the range of 400 nm to 490 nm.

- (g) (i) With reference to Table 1.2, explain why lamps radiating mainly violet and blue light are expected to increase the rate of photosynthesis in algae more than lamps radiating light of all wavelengths.

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 .....[2]

- (ii) A high rate of photosynthesis in the algal cells will result in high rate of coral growth.  
 Explain why.

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 .....[1]

Short term exposure to increased temperature can trigger a change in gene expression in corals.

For corals exposed to higher temperature, the expression of some proteins was compared against the control. Corals in the control group were exposed to normal temperature. The difference in gene expression between both groups of corals is shown in Table 1.3.

**Table 1.3**

| protein   | average concentration of protein / $\mu\text{M}$ |                                      |
|-----------|--------------------------------------------------|--------------------------------------|
|           | control group                                    | corals exposed to higher temperature |
| BCL2      | 16                                               | 232                                  |
| cathepsin | 184                                              | 10                                   |
| HSP90     | 20                                               | 149                                  |

- (h) (i) Calculate the percentage increase in expression of BCL2 for corals exposed to higher temperature.

[1]

- (ii) Using Table 1.3, describe the effect of higher temperature on the expression of HSP90.

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 .....[1]

- (iii) Based on your knowledge of gene expression, suggest how the expression of HSP90 in corals exposed to higher temperature can be regulated at the transcriptional level.

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 ..... [2]

[Total: 30]

- 2 Fig. 2.1 shows some organelles in a mammalian liver cell.

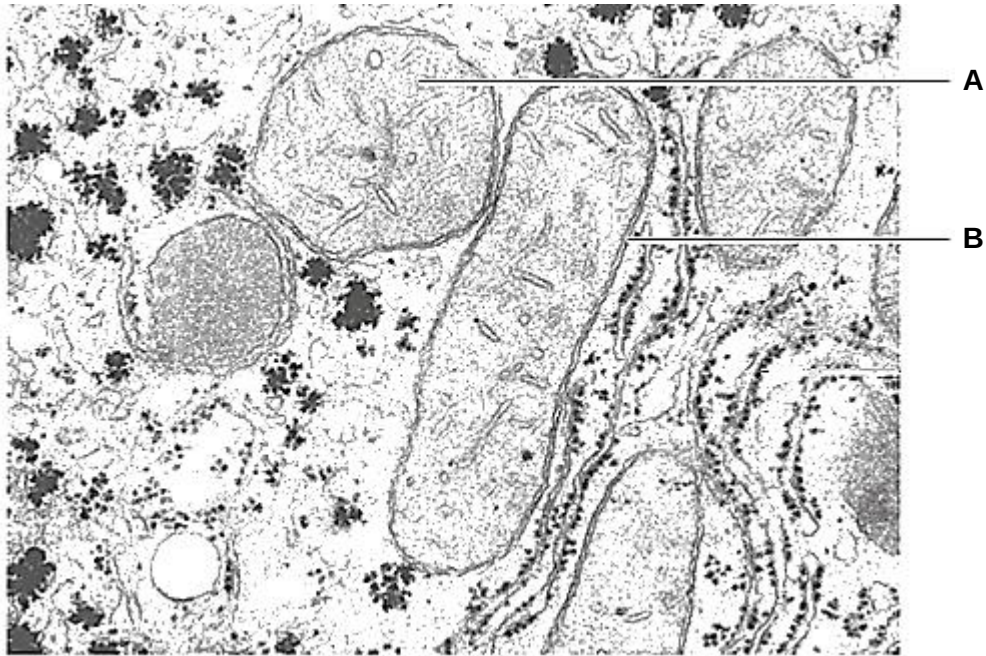


Fig. 2.1

- (a) With reference to Fig. 2.1, explain if **A** and **B** are the same organelle.

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..... [2]

- (b) During maturation, a red blood cell loses its nucleus and most other organelles, such as the ribosomes, rough and smooth endoplasmic reticulum, lysosomes and mitochondria. This gives more space for haemoglobin molecules.

Although there are no mitochondria, a mature red blood cell is still able to synthesise ATP.

Explain how a mature red blood cell can continue to synthesise ATP.

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..... [3]

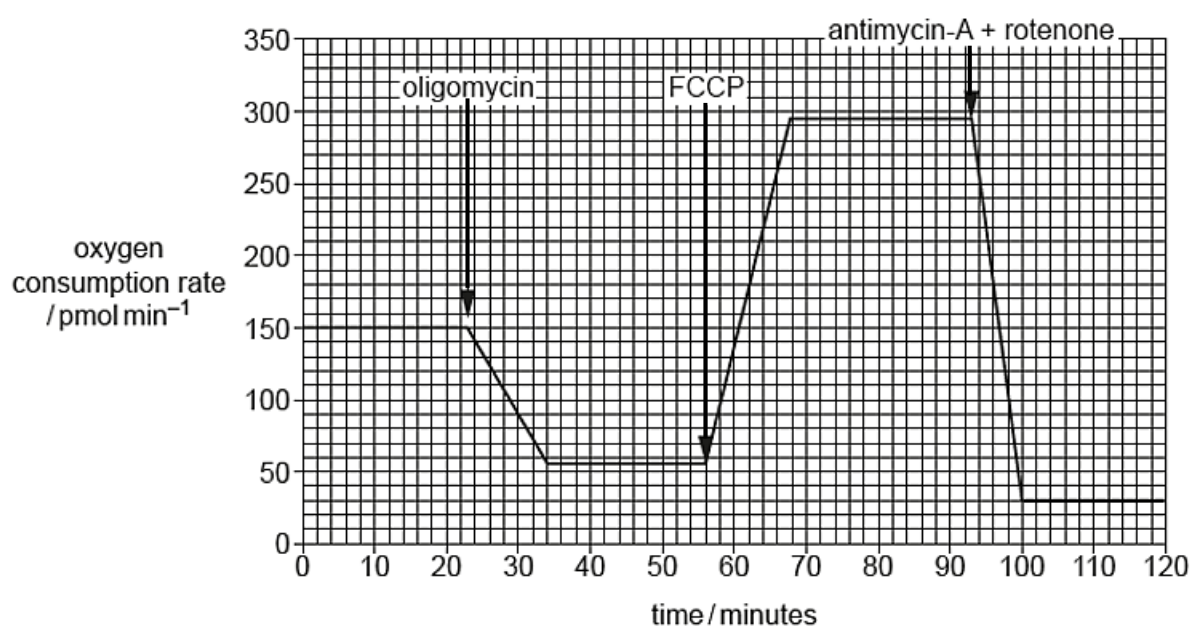
- (c) A number of chemicals can disrupt respiration. Some of these are listed in Table 2.1.

**Table 2.1**

| chemical               | effect on mitochondrion                                                                                                                         |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| oligomycin             | inhibits ATP synthase, thereby leading to steep proton gradient across the cristae and inhibition of active transport of protons across cristae |
| FCCP                   | causes protons to leak out of the intermembrane space back into the mitochondrial matrix without going through ATP synthase                     |
| antimycin-A + rotenone | inhibits the electron transport chain                                                                                                           |

Scientists measured the oxygen consumption rate (OCR) of some cells. They then investigated how the OCR changed when oligomycin was added at 23 minutes, followed by FCCP at 56 minutes, followed by antimycin-A + rotenone at 93 minutes.

Fig. 2.2 shows the results obtained.



**Fig. 2.2**

- (i) Explain each of the following two statements with reference to events at the inner mitochondrial membrane:

Oligomycin causes a drop in OCR

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..... [2]

FCCP causes a rise in OCR

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..... [2]

- (ii) During respiration, protons 'leak' through the cristae into the mitochondrial matrix. The OCR as a result of this leakage can be calculated by determining the difference between the OCR 20 minutes after the addition of oligomycin and the OCR 20 minutes after the addition of antimycin-A + rotenone.

Calculate the OCR caused by proton leakage. Show your working clearly.

..... $\text{pmol min}^{-1}$

[1]

[Total: 10]

- 3 Down's syndrome may be caused by a trisomy of chromosome 21, but may also result from the fusion of a large portion of chromosome 21 with chromosome 14, forming a single chromosome 14-21. The remaining portion of chromosome 21 is lost, but has no effect on the individual.

Fig. 3.1 shows the karyotype of a diploid ( $2n$ ) testis cell from a phenotypically normal individual carrying a 14-21 fusion chromosome, indicated with an arrow. The centromere of each chromosome is positioned on the dotted line. This individual has a higher chance of having a child with Down's syndrome.

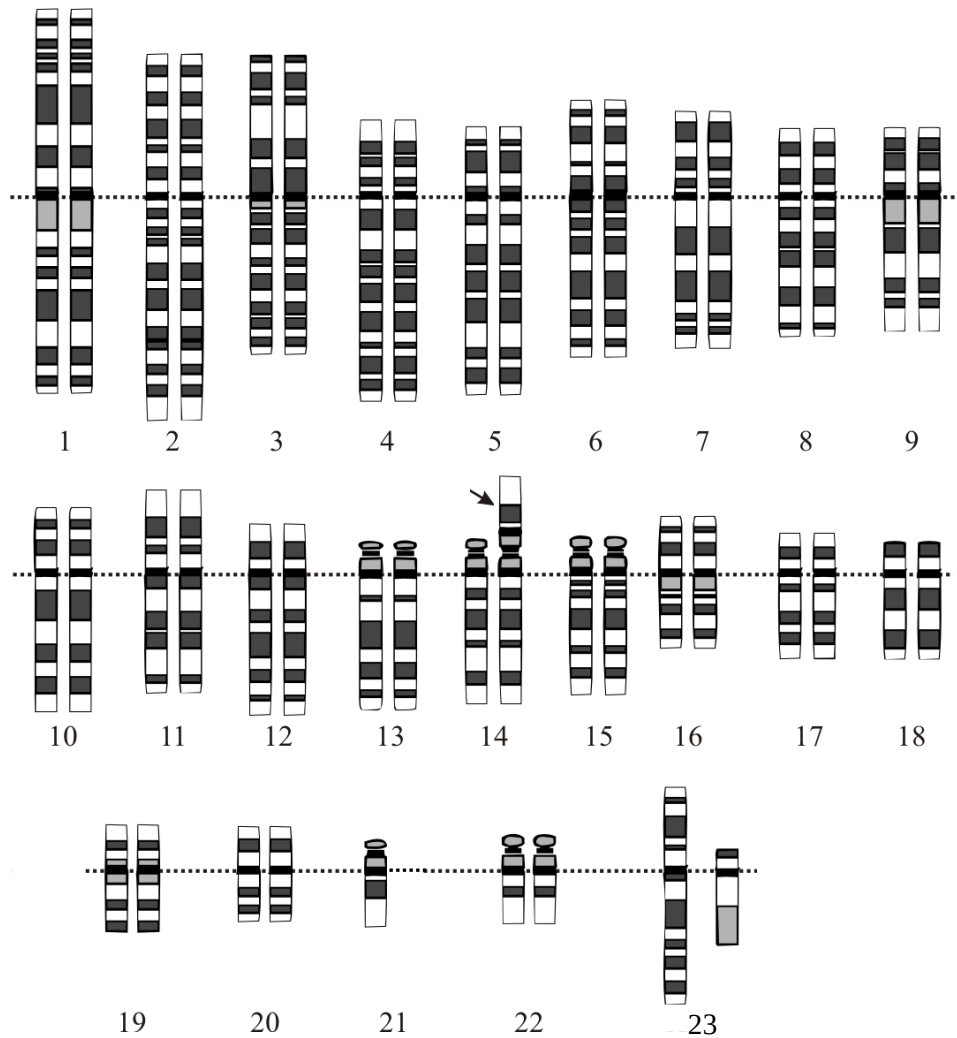


Fig. 3.1

- (a) (i) Suggest the type of chromosomal mutation that has occurred in this individual.

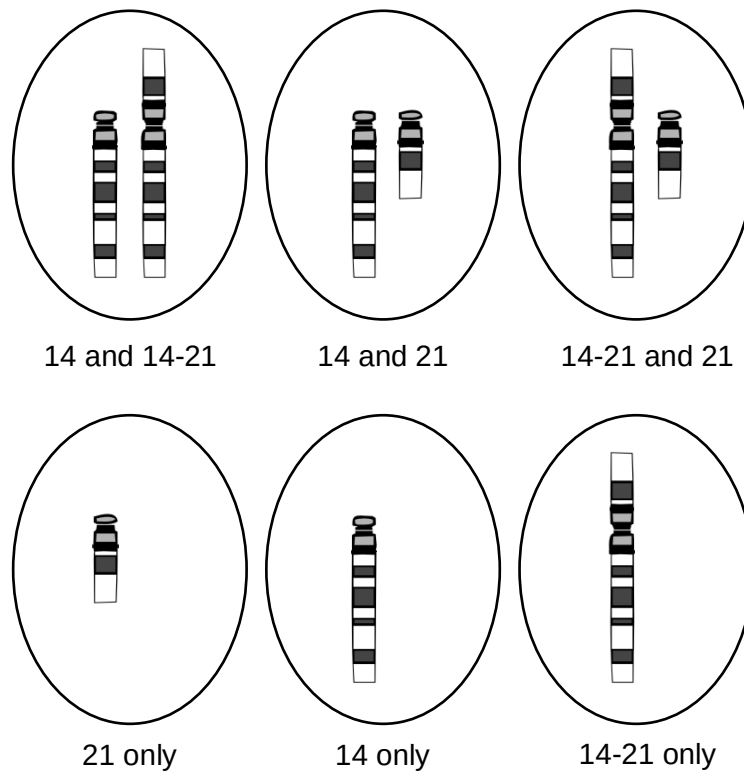
..... [1]

- (ii) State and explain **one other** phenotypic feature of this individual.

.....  
 ..... [1]



- (b) Fig. 3.2 shows all possible combinations of chromosomes 14, 21, and 14-21 in the gametes produced by the same individual whose karyotype is shown in Fig. 3.1.



**Fig. 3.2**

It is known that monosomy-14, monosomy-21, and trisomy-14 are all lethal conditions, and the embryo will not develop beyond 20 weeks.

- (i) Assuming that a new-born child inherits a normal gamete from the other parent, state the probability that the child is phenotypically normal.

probability = ..... [1]

- (ii) Explain your answer in (b)(i).

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..... [3]

Upon fertilisation, a zygote contains two copies of chromosome 21 and a fusion chromosome 14-21. All the daughter cells that arise from this zygote have trisomy-21.

(iii) Explain how events in the cell cycle results in all daughter cells having trisomy-21.

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..... [4]

[Total: 10]

**Section B**

Answer **one** question in this section.

Write your answers in the separate Answer Booklet provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)**, and **(b)** etc., as indicated in the question.

- 4 (a)** Using named examples, explain the importance of having a three-dimensional structure in different biological polymers. [15]

- (b)** *Shigella* bacteria can cause diarrhoea, fever, abdominal pain. Antibiotics such Ciprofloxacin help treat patients with severe infection or weakened immune system, and reduce the spread of infection by decreasing the number of days the patient has diarrhoea.

However, the number of *Shigella* infections have increased notably since 2013. These strains exhibit resistance to Ciprofloxacin.

Explain how Ciprofloxacin-resistant strains of *Shigella* could have emerged from a single strain of *Shigella*. [10]

**OR**

- 5 (a)** Discuss the causes and significance of variation and diversity in organisms. [15]

- (b)** The human body is capable of responding to changes in the environment. For example, secretion of digestive juices, such as saliva and intestinal juice, occurs in response to ingestion of food. These juices contain enzymes that break down complex carbohydrates into simpler sugars that can be absorbed into the blood.

Describe the cellular changes that occur in response to a meal rich in complex carbohydrates. [10]

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